

Conjecture 00000001128 is false:
 $M(321, 3) = 1$ is not a multiple of $\min(3, 3, 2)! = 2$

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1 The conjecture

Conjecture 00000001128 (English text, verbatim): “Definition: $M(w, j)$ is the maximal monomial coefficient in the degree- j part of the Schubert polynomial $\mathfrak{S}_w$. Conjecture: $M(w, j)$ equals $\min(j, \ell(w), \lceil \ell(w)/2 \rceil)!$ times a lattice-path count $C(w)$, where $C(w)$ has a product formula on 321-avoiding permutations; in particular $M(w, \ell(w))$ is an explicit function of the increasing-subsequence count of w .” The Chinese text says the same.

Reading. The main clause asserts: there is a function C (a lattice-path count, so a nonnegative integer) with

$$M(w, j) = \min(j, \ell(w), \lceil \ell(w)/2 \rceil)! \cdot C(w) \quad (*)$$

for all permutations w and degrees j . We refute $(*)$ already at $j = \ell(w)$ (the case the text singles out), even allowing $C(w)$ to be any integer. Two readings are covered: (R1) w ranges over all permutations; (R2) $(*)$ is only claimed for 321-avoiding w . The clauses about a product formula and about increasing subsequences are not used.

2 Definitions

Variables are 0-indexed: $x_0, \dots, x_n$ for S_{n+1} (the text’s $x_1, \dots, x_{n+1}$). $s_i = (i, i+1)$; ws_i is w with positions $i, i+1$ swapped in one-line notation; $\ell(w) = \#\{(a, b) : a < b, w(a) > w(b)\}$ (inversions); w_0 is the longest permutation $k \mapsto n - k$.

Definition 1 (divided difference). $\partial_i P = (P - s_i P)/(x_i - x_{i+1})$, where s_i swaps x_i and x_{i+1} .

Definition 2 (Schubert polynomials, Lascoux–Schutzenberger). $\mathfrak{S}_{w_0} = x_0^n x_1^{n-1} \cdots x_{n-1}$; $\partial_i \mathfrak{S}_w = \mathfrak{S}_{ws_i}$ if $w(i) > w(i+1)$, and $\partial_i \mathfrak{S}_w = 0$ if $w(i) < w(i+1)$.

Source (Wikipedia, “Schubert polynomial”, retrieved): “ $\mathfrak{S}_{w_0} = x_1^{n-1} x_2^{n-2} \cdots x_{n-1}^1$ ”, “ $\partial_i \mathfrak{S}_w = \mathfrak{S}_{ws_i}$ ” if “ $w(i) > w(i+1)$ ”, where ∂_i sends P to “ $(P - s_i P)/(x_i - x_{i+1})$ ”; “Schubert polynomials can be calculated recursively from these two properties.”

Definition 3. $M(P, j)$ is the maximum of the coefficients of the monomials of total degree j in the support of P (Lean: `WithBot ℤ`, $\perp$ if there is none).

3 Proof

Lemma 4. $x_i - x_{i+1}$ divides $P - s_i P$ for every $P \in \mathbb{Z}[x_0, \dots, x_n]$, and the quotient is unique.

Proof. Induction on P : constants give 0; sums are clear; for $P \cdot x_k$, $Px_k - s_i(P)x_{s_i(k)} = (P - s_i P)x_k + s_i(P)(x_k - x_{s_i(k)})$, and $x_k - x_{s_i(k)}$ is 0 or $\pm(x_i - x_{i+1})$. Uniqueness: $\mathbb{Z}[x]$ is a domain and $x_i \neq x_{i+1}$. $\square$

Theorem 5 (R1). In S_3 , $w = 321 = w_0$, $\ell(w) = 3$ and $\mathfrak{S}_{321} = x_0^2 x_1$, so $M(321, 3) = 1$. The right side of $(*)$ is $\min(3, 3, 2)! \cdot C = 2C$, and $1 = 2C$ has no integer solution.

Proof. $\mathfrak{S}_{321} = \mathfrak{S}_{w_0} = x_0^2 x_1$ is the base case of the recursion. Its only monomial has degree 3 and coefficient 1. The inversions of 321 are $(0, 1), (0, 2), (1, 2)$. $\square$

To show the recursion is consistent (so the statement is not vacuous), the six polynomials $\mathfrak{S}_{123} = 1$, $\mathfrak{S}_{213} = x_0$, $\mathfrak{S}_{132} = x_0 + x_1$, $\mathfrak{S}_{231} = x_0 x_1$, $\mathfrak{S}_{312} = x_0^2$, $\mathfrak{S}_{321} = x_0^2 x_1$ are checked against all twelve conditions (w, i) , each by an explicit identity $(x_i - x_{i+1})Q = P - s_i P$.

Theorem 6 (R2). $w = 4123 \in S_4$ avoids 321, $\ell(w) = 3$ and $\mathfrak{S}_{4123} = x_0^3$, so $M(4123, 3) = 1 \neq 2C$.

Proof. $4321 \rightarrow 4312 \rightarrow 4132 \rightarrow 4123$ is a chain of descent steps at positions 2, 1, 2 (0-indexed), so $\mathfrak{S}_{4123} = \partial_2 \partial_1 \partial_2 (x_0^3 x_1^2 x_2)$ for any family satisfying the characterization. We give all 24 Schubert polynomials of S_4 explicitly and check all 24×3 conditions; this family has $\mathfrak{S}_{4123} = x_0^3$, so every family does. $\square$

For reference, every Schubert polynomial of S_4 has maximal coefficient 1, while $[\ell/2]! \geq 2$ once $\ell \geq 3$; so $(*)$ fails for all 15 elements of S_4 of length ≥ 3 (computed, not used).

4 Lean formalization

File `lean/Conjecture1128/Basic.lean` (347 lines, only import `Mathlib`). Objects: `s i = swap i.castSucc i.succ`; `sAct i P = rename (s i) P`; `dd i P`, the quotient given by `dvd_sub_sAct` (divisibility for every P), with `dd_spec` and uniqueness `dd_eq`; `xDelta n =  $\prod_k x_k^{n-k}$` ; `IsSchubert S` (fields `top`, `step`, `zero`: exactly the characterization above with $w_0 = \text{Mathlib's } \text{Fin.revPerm}$); `len` (inversions); `maxCoeff P j`; `formula l j C =  $\min(j, \min(l, (l+1)/2))! \cdot C$` (in $\mathbb{N}$, $(l+1)/2 = \lceil l/2 \rceil$). Main theorems:

- `disproof: IsSchubert S3`, and for every S with `IsSchubert S` there is no $C : \text{Perm (Fin 3)} \rightarrow \text{Int}$ with `maxCoeff (S w) j = formula (len w) j (C w)` for all w , $j = \text{len } w$. It uses `disproof_321`, `schubert_321`, `len_w321`, `maxCoeff_x0sq_x1`, `isSchubert_S3`.
- `disproof_321_avoiding: IsSchubert S4, Avoids321 w4123`, and for every S with `IsSchubert S` no integer C satisfies $(*)$ at $(4123, \ell(4123))$. It uses `schubert_4123`, `isSchubert_S4`, `w4123_facts`, `maxCoeff_x0cube`.

`#print axioms` for both gives `[propext, Classical.choice, Quot.sound]`; no `sorry`, no `native_decide`. `decide` is used only on permutations of `Fin 3` and `Fin 4` (one-line values, inversions, 321-avoidance).

5 Relation to the earlier submission

An earlier submission (PR #201, orionsheep) had the right numbers but was closed. The reviewer wrote that it “hardcodes the polynomial as an opaque monomial list [(2,1,0,1)] with no divided-difference construction and no Lean proof that this list IS the Schubert polynomial of 321”, and asked for a version “deriving $\mathfrak{S}_{321}$ from w_0 by divided differences”. (An even earlier PR #161 formalized a wrong object.) This submission defines the divided difference as the exact quotient in $\mathbb{Z}[x_0, \dots, x_n]$, defines Schubert polynomials by the Lascoux–Schutzenberger characterization, and proves that the S_3 and S_4 families satisfy it. In S_3 , $321 = w_0$, so $\mathfrak{S}_{321} = x_0^2 x_1$ is the base case of the recursion; the S_4 case derives $\mathfrak{S}_{4123} = x_0^3$ from $w_0 = 4321$ by three divided differences and covers the 321-avoiding reading.

Source

Wikipedia, “Schubert polynomial”, https://en.wikipedia.org/wiki/Schubert_polynomial (formulas quoted above).